



Vincent Lannurien

ASSOCIATE PROFESSOR · LAB-STICC · SHAKER TEAM

Université de Bretagne Occidentale, UFR Sciences et Techniques

Département Informatique, 20 avenue Le Gorgeu, CS 83837, 29238 BREST CEDEX 3

☎ (+33) 6 75 46 83 43 | ✉ vincent.lannurien@univ-brest.fr | [in](#) [vincentlannurien](#) | [🎓 Scholar](#)

Interests

My research interests range from Embedded Software to the Serverless Cloud – systems that exhibit trade-offs between performance and availability in highly constrained, heterogeneous environments.

Education

ENSTA Bretagne

Brest, France

PHD IN COMPUTER SCIENCE

Oct. 2021 - Nov. 2024

- PhD thesis: *Dynamic allocation and placement on heterogeneous resources in a serverless cloud*.
- Director: Jalil Boukhobza. Co-director: Laurent d'Orazio. Co-advisors: Olivier Barais, Stéphane Paquetlet.

University of Western Brittany (UBO)

Brest, France

MASTER'S DEGREE IN EMBEDDED SOFTWARE ENGINEERING

2019 - 2021

- Ranked first in class.

University of Western Brittany (UBO)

Brest, France

BACHELORS'S DEGREE IN COMPUTER SCIENCE

2017 - 2019

- Ranked second in class.

University of Western Brittany (UBO)

Quimper and Brest, France

BACHELORS'S DEGREE IN ECONOMICS AND MANAGEMENT

2010 - 2013

Experience

University of Western Brittany

Brest, France

ASSOCIATE PROFESSOR

Sep. 2025 - Present

- Faculty of Sciences, Computer Science department;
- SHARP/SHAKER team, Lab-STICC, CNRS, UMR 6285.

ENSTA Bretagne

Brest, France

POSTDOCTORAL RESEARCHER

Jan. 2025 - Aug. 2025

- **LanguEdge**, Large Language Model (LLM) inference on heterogeneous edge platforms – characterization and scheduling for multi-tenant deployment:
 - Responsible for project planning and proposal writing: bibliographical study, research directions, budgeting;
 - Characterized LLM models with regards to generative quality, performance as well as energy consumption and carbon footprint.

ENSTA Bretagne

Brest, France

POSTDOCTORAL RESEARCHER

Oct. 2024 - Jan. 2025

- **DISPEED**, Intrusion Detection and Security/Performance/Energy Tradeoff for Drone Swarms:
 - Proposed a model encompassing all the project's contributions to present the system in the formalism of autonomous computing;
 - Wrote the final report, delivered to the Agence Innovation Défense (AID) in January 2025.

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Brest, France

RESEARCH ENGINEER & TEACHING ASSISTANT

Oct. 2021 - Oct. 2024

- Contributed in two projects within the institute (**SUPRA**, sovereign cloud project for sensitive applications; and **RPC**, private cellular networks);
- Joint work with the institute's Hardware Engineering team: characterization of heterogeneous hardware and software platforms based on performance and energy measurements;
- Presentation and valorization of our work at various milestones: meetings with industrial partners, biannual plenary sessions for both projects.

- As part of an embedded systems R&D team, charged of improving reliability in the software development process.
 - Shipped a new sensor driver development toolkit release, including a custom IDE, to industrial partners of the observatory lab.
 - Designed a simulator for the embedded HTTP daemon in order to speed up UI/UX development.
 - Devised and developed a server/client web application to allow remote compilation of the COSTOF2 firmware.

Service

CCGrid'24

Philadelphia, USA

SESSION CHAIR

May 2024

- Chaired two technical sessions.

EuroSys'24

Athens, Greece

ARTIFACTS EVALUATION COMMITTEE

April 2024

- Reviewed artifacts for three submissions.

Various venues

PROGRAM COMMITTEE

2022 - 2026

- Journal reviews: IEEE TSC (1), Elsevier FGCS (4), IEEE TC (1).
- Conference reviews: IEEE JCC'26 (4), IEEE/ACM CCGrid'24 (1), ACM/IEEE DAC'23 and '24 (2), DSD/SEAA '23 and '24 (2), MASCOTS '23 and '24 (2).
- Workshop reviews: SC/PDSW'25 (2), BPOD/BigData'22 (1).

Publications

JOURNAL ARTICLES

HeROsim: An Allocation and Scheduling Simulator for Evaluating Serverless Orchestration Policies

Vincent Lannurien, Laurent D'Orazio, Olivier Barais, Stéphane Paquelet, Jalil Boukhobza*IEEE Internet Computing* (2024): *Special Issue on Serverless Computing*. IEEE, 2024

CONFERENCE PROCEEDINGS

HeROcache: Storage-Aware Scheduling in Heterogeneous Serverless Edge - The Case of IDS

Vincent Lannurien, Camélia Slimani, Laurent D'Orazio, Olivier Barais, Stéphane Paquelet, Jalil Boukhobza*2024 IEEE/ACM 24th International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2024, Philadelphia, USA

HeROfake: Heterogeneous Resources Orchestration in a Serverless Cloud – An Application to Deepfake Detection

Vincent Lannurien, Laurent D'Orazio, Olivier Barais, Esther Bernard, Olivier Weppe, Laurent Beaulieu, Amine Kacete, Stéphane Paquelet, Jalil Boukhobza*2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2023, Bangalore, India

BOOK CHAPTERS

Serverless Cloud Computing: State of the Art and Challenges

Vincent Lannurien, Laurent D'Orazio, Olivier Barais, Jalil Boukhobza*Serverless Computing: Principles and Paradigms*, 2023, Springer, Cham

SOFTWARE

HeROsim is a serverless orchestration simulator, tracing allocation and scheduling events at the granularity of a user request, allowing to evaluate and compare various orchestration policies for serverless platforms in a private cloud.

Repository: <https://hal.science/hal-04468894>

Teachings

COURSES

Distributed Systems

University of Western Brittany

1ST YEAR MASTER'S DEGREE

2025

- Design and delivery of 50% of the (whole semester) unit: lectures, labs, project and exam.
- Graduate level, 60 students.
- Topics: OOP, design patterns, error handling, communications and authentication.

Software Engineering Application Project

University of Western Brittany

1ST YEAR MASTER'S DEGREE

2025

- Design of the project subject, supervision and evaluation of the students.
- Graduate level, 25 students.
- Topics: backend/frontend architecture, reactive programming, reliability.

Smart Objects and Robotics

University of Western Brittany

3RD YEAR BACHELOR'S DEGREE

2025

- Design and delivery of the (optional) unit: lectures, labs, and exam.
- Undergraduate level, 30 students.
- Topics: embedded programming, microcontrollers, sensors and actuators, control theory.

Introduction to Algorithms and Programming

University of Western Brittany

1ST YEAR BACHELOR'S DEGREE

2025

- Supervision of exercises and labs.
- Undergraduate level, 40 students.
- Topics: programming basics, data structures, algorithmic complexity.

Distributed and Virtualized Architectures

ENSTA

3RD YEAR ENGINEERING DEGREE

2021 - 2024

- Design and delivery of 25% of the (whole semester) unit: lectures, project and exam.
- Graduate level, 15 students.
- Topics: cloud-native development, asynchronous programming, performance metrics, cloud deployment.

Memory Technologies

ENSTA

3RD YEAR ENGINEERING DEGREE

2021 - 2024

- Preparation and supervision of lab exercises.
- Graduate level, 20 students.
- Topics: C programming, memory hierarchy, Linux I/O, performance measurements

Introduction to Linux Administration

ENSTA

2ND YEAR ENGINEERING DEGREE

2022 - 2024

- Design and delivery of the 16h unit: lectures, exercises and project.
- Graduate level, 15 students.
- Topics: shell scripting, networking, virtualization.

SQL Databases

ENSTA

1ST YEAR ENGINEERING DEGREE

2022

- Supervision of lab exercises (20h).
- Undergraduate level, 40 students.

STUDENT PROJECTS

Graduation projects

ENSIA (Algiers)

3RD YEAR ENGINEERING DEGREE

2025 - 2026

- Co-devised and supervised graduation project (1 year).
- Topics:
 - 2025 - 2026: **EdgeLLM**, *Optimizing LLMs for Edge Devices* (2 students).

Graduation projects

ESI (Algiers)

3RD YEAR ENGINEERING DEGREE

2023 - 2025

- Co-devised and supervised graduation projects (1 year).
- Topics:
 - 2024 - 2025: **MeSDGE**, *Metrology and performance characterization for distributed storage in embedded edge computing* (2 students).
 - 2023 - 2024: *Kernel-level I/O tracer design and caching policies for HPC applications* (1 student).

Systems Application

ENSTA

3RD YEAR ENGINEERING DEGREE

2022 - 2024

- Co-devised and supervised the research and development projects (6 months).
- Topics:
 - 2024: *Optimizing serverless resources allocation under carbon emissions constraints* (1 student)
 - 2023: *Web interface for visualization tools in a serverless orchestration simulator* (1 student)
 - 2023: *Private cloud resources provisioning: optimization using genetic algorithms* (1 student)
 - 2022: *Serverless execution trace parser and generator for heterogeneous workloads* (2 students)

Introduction to Scientific Research

ENSTA

1ST YEAR ENGINEERING DEGREE

2022

- Co-devised and supervised the research project (3 months).
- Topic:
 - *Cloud, datacenters and energy consumption: challenges and opportunities* (5 students)